Milestone 2 – Glucose_Explorers

Project goal

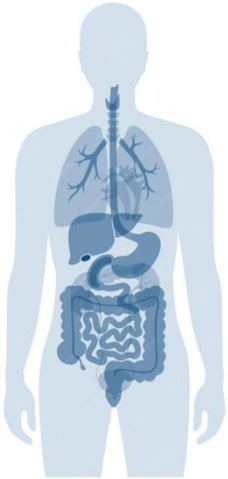
The goal of this project is to create an interactive educational website raising awareness about diabetes and its risk factors. By combining infographics with short explanatory texts, the website will allow users to build a clearer understanding of the disease, what it is, how it affects the body, and which lifestyle and medical factors are associated with it.

The website is organized around a central narrative: starting from a general introduction to diabetes and the clinical variables it influences, moving toward an exploration of individual risk factors, and ending with a personal, interactive tool that lets users situate themselves relative to the dataset population.

A possible enhancement would allow users to input their own lifestyle and simple medical history values and compare their profile to the 100,000-patient dataset through an interactive visualization.

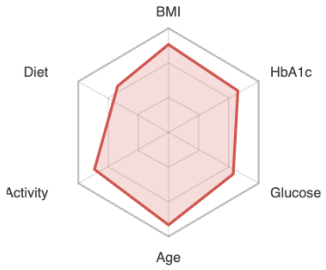
Principal visualizations

1 — The human body (core)



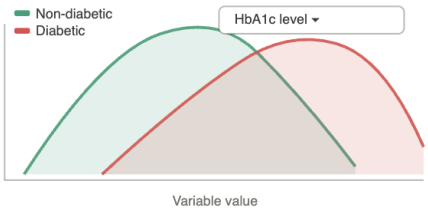
A human silhouette displaying the principal organs. The user can hover over each organ to reveal which clinical variables are related to it — for example, hovering over the pancreas shows its connection to insulin and glucose levels, while hovering over the heart surfaces cardiovascular risk indicators. This visualization serves as the educational entry point of the website, grounding abstract clinical variables in a physical, intuitive representation of the body.

2 — Interactive risk profile (core)



A spider (radar) chart where the user enters their own values via sliders — covering lifestyle variables such as screen time and physical activity, as well as simple medical history items such as hypertension. The chart updates in real time to show where the user sits relative to both the diabetic and non-diabetic populations in the dataset. The risk estimate is powered by a simple logistic regression model pre-trained on the dataset and exported as JavaScript coefficients, requiring no server-side computation.

3 — Distribution comparison (core)



A variable selector lets the user choose any clinical or lifestyle variable and see two overlapping distributions — one for diabetic patients, one for non-diabetic — rendered as smooth density curves. This gives an immediate visual sense of which variables separate the two groups most clearly, and how much overlap remains even for strong predictors like HbA1c or fasting glucose.

4 — Lifestyle impact shelf (enhancement)

A tabletop scene showing three everyday objects: a glass of wine, a dinner plate, and a pair of running shoes. Each object encodes one lifestyle variable and updates live as the user moves sliders — the glass fills or empties, the plate shifts from vegetables to processed food, the shoes change appearance with activity level. A thermometer alongside the objects responds to the combined effect of all inputs, rising and changing color to reflect the estimated impact on diabetes risk. This visualization makes the population-level patterns feel personal and tangible, without requiring any statistical literacy from the user.